ModelIt!

Teacher's Guide

Why Does Your Coffee Get Cold?

The Hidden Math of Thermal Energy Transfer

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-PS3-3, MS-PS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS3-3, MS-PS3-4

Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer. Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model demonstrating how temperature difference drives transfer rate and produces exponential decay cooling curves.
Disciplinary Core Idea	**PS3.A Definitions of Energy / PS3.B Conservation of Energy** — Temperature is a measure of the average kinetic energy of particles. Thermal energy transfers from regions of higher temperature to regions of lower temperature until equilibrium is reached.
Crosscutting Concept	**Cause and Effect** — Students identify that the temperature difference CAUSES the heat transfer, and as the effect (cooling) reduces the cause (temperature difference), the rate of change slows. This is a self-limiting system.

CAST Testing Alignment

MULTIPLE CHOICE:

A student heats two identical cups of water to 80 degrees C. Cup A is placed in a 10 degrees C room and Cup B is placed in a 30 degrees C room. After 10 minutes, which cup has lost more thermal energy, and why?

CONSTRUCTED RESPONSE:

Using Newton's Law of Cooling, explain why a cup of coffee left on a counter cools rapidly in the first 5 minutes but barely changes temperature in the last 5 minutes before reaching room temperature. Include a sketch of the cooling curve.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The cooling of a hot object is one of the most fundamental and observable physics phenomena. Newton's Law of Cooling describes the exponential decay pattern, and this same mathematical pattern appears in radioactive decay, capacitor discharge, population decline, and countless other systems.

Newton's Law of Cooling

Isaac Newton discovered that the rate of heat loss from an object is proportional to the temperature difference between the object and its surroundings. This means that a 90-degree C object in a 20-degree C room loses heat much faster than a 40-degree C object in the same room. The bigger the gap, the faster the flow.

Exponential Decay

The cooling curve follows an exponential decay pattern. Temperature drops rapidly at first (when the gap is large), then progressively slows as the gap shrinks. Mathematically, the temperature approaches room temperature but never quite reaches it (in theory). In practice, the difference becomes so small that it is undetectable.

Thermal Equilibrium

When two objects reach the same temperature, they are in thermal equilibrium. Net heat transfer is zero because there is no temperature difference to drive it. This is the Second Law of Thermodynamics in action: energy naturally disperses from concentrated (hot) to diffuse (spread out), never the reverse without external energy.

Real-World Applications

This principle governs building insulation (slowing heat transfer through walls), cooking (heat must transfer from the oven to the center of food), climate science (Earth radiates heat to space based on temperature difference), and medicine (hypothermia treatment, fever management). Engineers use these principles daily.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify hot object temperature, room temperature, temperature difference, transfer rate, and object temperature over time as the key components of the thermal system.

E **Establish Relationships**
Students determine that temperature difference positively drives transfer rate, transfer rate negatively affects object temperature, and as the object cools, temperature difference shrinks, creating a self-slowing system.

V **Visualize & Model**
Students build a model showing how the interplay between temperature difference and transfer rate produces the characteristic exponential decay cooling curve.

E **Evaluate Outcomes**
Students run hot-in-cold-room vs. hot-in-warm-room scenarios and compare cooling curves to observe how room temperature affects the equilibrium point and cooling speed.

R **Revise & Extend**
Students add insulation level or object mass to explore how additional variables modify the cooling curve shape.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You make the perfect cup of coffee. You take one phone call. It is cold. Where did all that heat go?"*

Watch For

If possible, bring a hot cup of water and a thermometer. Show the temperature dropping in real time.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to discover the hidden math behind every cooling curve in the universe."*

Watch For

Have students read objectives. Pre-teach 'thermal equilibrium' and 'exponential decay.'

Slide 3: Big Question

3 min

Say: *"Your coffee cools fast at first, then slows to a crawl. That is NOT random. There is an elegant law behind it."*

Watch For

Have students sketch their prediction of a cooling curve. Collect and display several predictions.

Slide 4: LEVER Framework

2 min

Say: *"We will build a model to discover Newton's Law of Cooling through simulation."*

Watch For

Preview the 5 components. Emphasize that this model has a self-limiting feedback mechanism.

Slide 5: Activity 1: Components

8 min

Say: *"What controls how fast heat transfers? The temperature of the object? The room? Both?"*

Watch For

Guide sorting. Discuss why both temperatures are external inputs the student controls.

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the key insight: as the object cools, the temperature difference SHRINKS, which slows the transfer rate. The system slows itself down."*

Watch For

Draw the feedback loop on the board. Show how the output (cooling) reduces the input (temperature difference).

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare coffee in a cold room versus coffee in a warm room. Which cools faster?"*

Watch For

Students predict curves first. Run both scenarios. Overlay the curves to compare.

Slide 8: Discoveries

5 min

Say: *"This exponential decay pattern is everywhere. Cooling coffee, draining batteries, radioactive decay. Same math, different systems."*

Watch For

Show examples of exponential decay in other domains. Discuss thermal equilibrium.

Slide 9: STEM Challenge

12 min

Say: *"Now you are biomedical engineers. Design a system that keeps medicine at body temperature for 4 hours."*

Watch For

Distribute materials and constraints. Groups design, test, and measure passive thermal systems.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Hot Object Temperature, Room Temperature
- **INTERNAL:** Temperature Difference, Transfer Rate, Object Temperature Over Time

Relationships

- **Hot Object Temperature to Temperature Difference:** POSITIVE (+) — A hotter starting object creates a larger temperature gap with the room, which drives faster initial heat transfer.
- **Room Temperature to Temperature Difference:** NEGATIVE (-) — A warmer room means a smaller gap between the object and its surroundings, reducing the driving force for heat transfer.
- **Temperature Difference to Transfer Rate:** POSITIVE (+) — The larger the temperature gap, the faster heat flows. This is Newton's Law of Cooling: transfer rate is proportional to temperature difference.
- **Transfer Rate to Object Temperature Over Time:** NEGATIVE (-) — A faster transfer rate causes the object to cool more quickly. As the object cools, temperature difference shrinks, which slows the transfer rate. This creates the exponential decay curve.

Key Discoveries

- Heat transfer rate is proportional to the temperature difference. Bigger gap means faster transfer
- The cooling curve is NOT a straight line. It is an exponential decay curve that flattens as the object approaches room temperature
- A hot object in a warmer room cools more slowly because the temperature difference driving the transfer is smaller
- The object can never cool below room temperature because at equilibrium, the temperature difference is zero and transfer stops
- This same pattern appears everywhere: cooling drinks, warming cold objects, climate systems, and even electric circuits

Multiple Choice Post-Assessment Answer Key

1. A cup of hot chocolate at 85 degrees C is placed in a 20 degrees C room. After 10 minutes, the temperature is 50 degrees C. After 20 more minutes, the temperature is 28 degrees C. What explains this pattern?

Correct: B) The drink cooled faster at first because the large temperature difference drove rapid heat transfer, then slowed as the gap shrank

Correct! This is Newton's Law of Cooling in action. The initial 65-degree gap drove fast cooling (35 degrees in 10 minutes). As the gap shrank to 30 degrees, the transfer rate slowed (only 22 degrees in 20 minutes). The driving force for heat transfer weakens as the temperatures converge.

2. In the thermal energy model, why can the hot object never cool below room temperature?

Correct: B) At room temperature, the temperature difference is zero, so there is no driving force for further heat transfer

Correct! Heat transfer requires a temperature difference to drive it. When the object reaches room temperature, the difference is zero and the transfer rate drops to zero. With no driving force, no more heat can flow. This is thermal equilibrium.

3. Two identical cups of 90 degrees C water are placed in different rooms. Cup A is in a 10 degrees C room and Cup B is in a 30 degrees C room. After 15 minutes, which statement is true?

Correct: B) Cup A lost more heat because the larger temperature difference drove faster transfer

Correct! Cup A has a temperature difference of 80 degrees (90 minus 10) while Cup B has only 60 degrees (90 minus 30). The larger difference drives faster heat transfer, so Cup A cools more quickly and loses more thermal energy in the same time period.

4. Which graph shape correctly represents how a hot object's temperature changes over time as it cools toward room temperature?

Correct: B) A curve that drops steeply at first, then gradually flattens as it approaches room temperature

Correct! The cooling curve is an exponential decay. It drops steeply when the temperature difference is large (driving fast transfer), then flattens as the difference shrinks and transfer slows. The curve approaches but never quite reaches room temperature.

5. A thermos keeps coffee hot for hours while a metal cup lets coffee cool in minutes. Based on the thermal energy model, what does the thermos change?

Correct: C) The thermos slows the transfer rate by reducing how quickly heat can flow through the container walls

Correct! A thermos does not change the temperature difference or create heat. It adds insulation that dramatically slows the RATE at which heat transfers through the walls. The same driving force (temperature difference) is present, but the transfer rate is reduced by the insulating layers.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Objects cool at a constant rate"	Cooling is NOT linear. Objects cool fast at first (large temperature gap drives fast transfer) and slow down as they approach room temperature. The curve is exponential, not a straight line. This is a fundamental pattern in physics.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Cold flows into hot objects"	There is no such thing as 'cold' flowing. Only thermal energy (heat) moves, and it always moves from hot to cold. When your coffee cools, heat is leaving. When your hands feel cold touching metal, your body heat is flowing INTO the metal, not cold flowing out.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Room temperature is a property of the object"	Room temperature is the temperature of the environment, not the object. The object approaches room temperature through heat transfer. Different rooms have different temperatures. The same coffee would reach different final temperatures in different rooms.